

Probability and data science...

... before data science even existed.

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Part I: Ronald Fisher

What did he do?

A bit of everything!

- **Biology:** Integrate the work of Mendel and Darwin.
- **Economics:** Develop the "Fisher Index" for measuring inflation.
- **Statistics:** Invent the whole field!



The father of statistics

- **Maximum likelihood:** Best way guess the "truth" from small samples.
- **Real-world impact:** Allied statisticians used to solve the *German Tank Problem* in WWII, estimating enemy production purely from captured serial numbers.
- **Experimental design:** Proved that experiments must be *randomized* to eliminate hidden variables.

The Lady Tasting Tea

- **The scene:** Afternoon tea, Cambridge, 1919.
- **The claim:** Muriel Bristol asserts she can taste whether milk or tea was poured into the cup first.
- **The reaction:** Fisher is skeptical. He designs an experiment on the spot.



The Protocol

- **The design:** 8 *indistinguishable* cups total (4 milk-first, 4 tea-first).
- **The challenge:** Identify the 4 milk-first cups.
- **The math:** $\binom{8}{4} = 70$ possible combinations.
- **Pure luck:** Probability of guessing perfectly is $1/70 \approx 0.014$ (1.4%).

The Null-Hypothesis (H_0)

- **Definition:** The baseline assumption of *no effect* or *no skill*.
- **In this case:** " H_0 : Muriel is guessing at random."
- **Quote:** "*The null hypothesis is never proved or established, but is possibly disproved, in the course of experimentation.*"

The skeptical judge

- H_0 is the foundation of the scientific method.
- We default to the null unless evidence is overwhelming.
- Prevents scientists from seeing "patterns" in random noise.
- Any new discovery requires overwhelming evidence.

The p-value

- **Definition:** The probability of seeing results *at least as extreme* as ours, assuming H_0 is true.
- A mathematical measure of "Surprise."
- $p = P(\text{Data} \mid H_0)$

The $p < 0.05$ Threshold

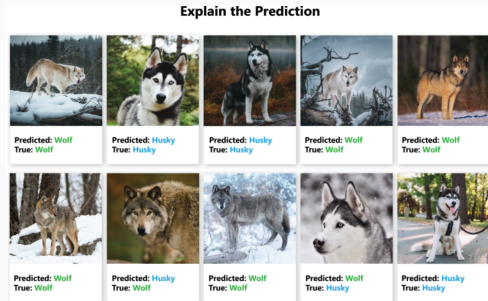
- Fisher suggested 1 in 20 (5%) as a "convenient" threshold for statistical significance.
- **Crucial Warning:** $p < 0.05$ does *not* mean the theory is 95% true. It means the data is rare under the Null.
- This arbitrary limit still governs modern medicine and science today.

The Verdict on the Tea

- **The Result:** Muriel Bristol identified all cups correctly.
- **The Math:** $p = 0.014$. Since $0.014 < 0.05$, Fisher rejected H_0 .
- **The Chemistry:** Pouring hot tea into cold milk preserves proteins. Milk into hot tea scalds it. She was right.

Fisher and AI (The Husky/Wolf Problem)

- AI is often fed huge data piles without Fisher's rigorous "controlled design."
- Lack of randomization might lead to wrong conclusions!.



Part II: Thomas Bayes

1. Who is Thomas Bayes?

- 18th-century Presbyterian minister and philosopher.
- **Inverse Probability:** Calculating the probability of a cause based on its effects.
- **Quote (via Richard Price):** *"Given the number of times in which an unknown event has happened and failed: Required the chance that the probability of its happening in a single trial lies somewhere between any two degrees of probability."*



2. The Bayesian Revolution

- **Fisher:** $P(\text{Data} \mid \text{Theory})$
- **Bayes:** $P(\text{Theory} \mid \text{Data})$
- Bayes provides the mathematics of *changing your mind* based on new evidence.

3. Thinking Like a Bayesian

- **The Prior:** Your belief *before* seeing evidence.
- **Scenario:** A loud "thump" on your roof. Plane crash or tree branch?
- Your brain uses Priors to instantly dismiss the plane crash. The baseline probability is simply too low.

4. The Bayes Formula

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

- $P(A | B)$: **Posterior** (Probability the theory is true given the data).
- $P(B | A)$: **Likelihood** (How well the data fits the theory).
- $P(A)$: **Prior** (Your initial belief).
- $P(B)$: The total baseline probability of observing this data across all possible theories (hard to compute!).
- *"Extraordinary claims require extraordinary evidence."*

5. Spam Filtering (Bayes in Action)

- **Prior:** Your historical overall spam rate.
- **Evidence:** Email contains "Winner" and "Wire Transfer."
- **Adaptation:** Learns continuously. Flagging a "crypto" email increases the Prior for that keyword in your specific inbox.

6. Medical False Positives (The Trap)

- **Scenario:** Disease affects 1 in 10,000. Test is 99% accurate.
- **Result:** You test positive.
- **The Math:** You intuitively think you have a 99% chance of being sick. Bayesian math proves it's only $\approx 1\%$.
- **Why?** The prior (1 in 10,000) is so low that the 1% error rate on healthy people swamps the true positives.

7. Polling is Bayesian Inference

- **The Fisher View:** $P(\text{Poll Data} \mid \text{True Vote})$. "If the candidate is winning, how likely is this poll?"
- **The Bayesian View:** $P(\text{True Vote} \mid \text{Poll Data})$. "Given this poll, what is the probability the candidate is winning?"
- In polling, the **Prior** is the "Demographic Model" (who the pollster thinks will actually show up on Election Day).

8. The 2016 Polling Anomaly: Rigid Priors

- **The Prior:** Most 2016 pollsters assumed the electorate would look exactly like the 2012 Census.
- **The Weighting Trap:** If a pollster needed 10% young voters but only 1% answered the phone, they *weighted* that 1% up by 10x.
- **The Result:** The "prior" (the model) became so heavy it effectively ignored the "evidence" (the raw data) coming from the field.

9. The One-Man Demographic

- **The outlier:** One specific USC/L.A. Times poll showed Trump winning because of a single 19-year-old Black man in Illinois.
- **The effect:** Because he was the *only* person in his demographic block to respond, his single vote was weighted **30 times higher** than average.
- **The lesson:** When your prior (weighting) is too rigid, a single piece of anomalous evidence is amplified into a false reality.

9. Bayes and Modern AI

- **Weights as priors:** Just like the L.A. Times poll forced mathematical "weights" to shape reality, Deep Learning trains by updating billions of Bayesian weights/priors.
- **Fine-tuning:** Developers give AI a strong Prior to act as a "helpful assistant."
- **Hallucinations:** If an LLM's prior for a narrative or sentence structure is too rigidly weighted, it completely ignores the user's evidence and confidently outputs a plausible lie.

Part III: Stanislaw Ulam

1. Who is Stanislaw Ulam?

- Polish-American mathematician on the Manhattan Project.
- Co-inventor of the Teller-Ulam Hydrogen Bomb design.
- Real-world Bayesian systems (like nuclear reactions) have thousands of variables. The math becomes impossible to solve on paper.



2. Playing Solitaire

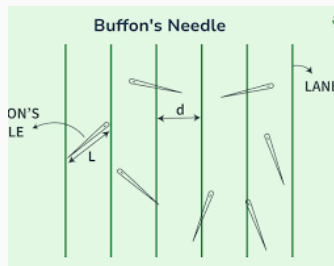
- **1946:** Recovering from brain surgery, playing Solitaire.
- **Quote:** *"What are the chances that a Canfield solitaire laid out with 52 cards will come out successfully? After spending a lot of time trying to estimate them by pure combinatorial calculations, I wondered whether a more practical method... might not be to lay it out say one hundred times and simply observe and count..."*

3. The Monte Carlo Method

- **The concept:** Use controlled randomness to approximate deterministic truths.
- Named after the Monaco casino where his uncle gambled.
- **The bridge:** When Bayes' equations are too complex, we use Ulam's simulations to approximate the Posterior.

4. Buffon's Needle

- Toss needles of length L randomly onto slatted wooden floors of width d
- **The math:** The probability of crossing a line is $\frac{2L}{\pi d}$.
- Therefore, $\pi \approx \frac{2L}{d} \times \frac{\text{Total Tosses}}{\text{Line Crosses}}$.
- **Human bias:** Lazzarini (1901) famously obtained 1808 crossings out of 3408 tosses with $L = \frac{5}{6}D$.
- This is "half luck, half cheat", and gives $\pi \approx 355/113$.



Modern Implications: Weaponizing MCMC

- **The setup:** In modern elections, data scientists use MCMC to generate millions of random, legal voting maps to see what a "fair" distribution looks like.
- **The evidence:** In recent Supreme Court cases, mathematicians used MCMC to prove that certain state maps were 1-in-24,000 extreme outliers. They were mathematically rigged.
- **The malicious trap:** While political operatives were using the *exact same MCMC algorithms*, the unfairness came from the stopping rule.

Closing Summary

- **Fisher:** Established rigorous experimental design and the skeptical judge (H_0).
- **Bayes:** Built the logical framework for updating beliefs with new evidence.
- **Ulam:** Proved that when the math is too complex, we can approximate the truth through random simulation.

Thank You